INCH-POUND

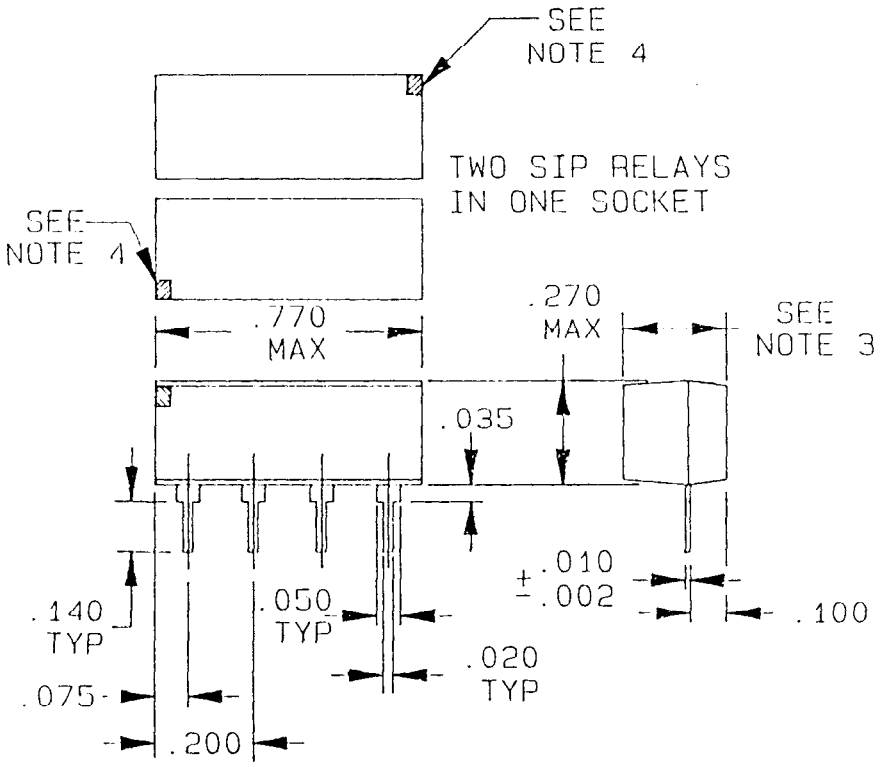
MIL-R-83516/3C(USAF)
30 September 1990
SUPERSEDING
MIL-R-83516/3B(USAF)
3 March 1983

MILITARY SPECIFICATION SHEET

RELAYS, REED, DRY, SINGLE IN-LINE PACKAGE (SIP), GENERAL PURPOSE

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and and Standards (DODISS) specified in the solicitation: MIL-R-83516.

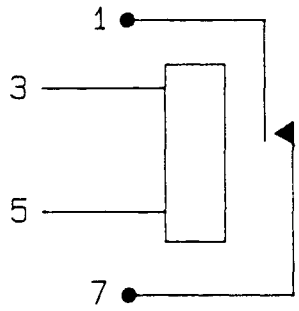


© FIGURE 1. Outline drawing.

© denotes changes

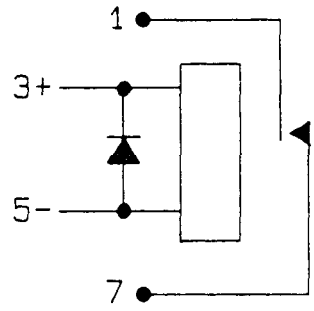
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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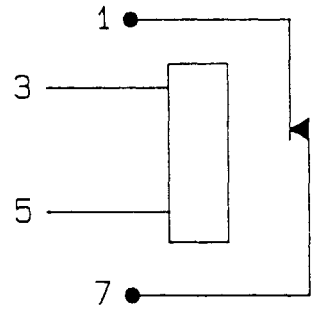
CIRCUIT DIAGRAM A

FORM A



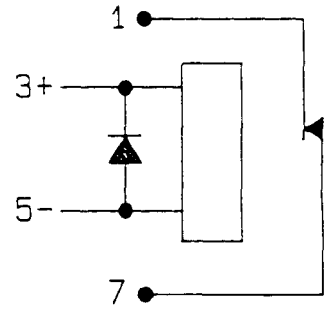
CIRCUIT DIAGRAM B

FORM A



CIRCUIT DIAGRAM C

FORM B



CIRCUIT DIAGRAM D

FORM B

Inches	mm		Inches	mm
.002	0.05		.075	1.90
.010	0.25		.100	2.54
.020	0.51		.140	3.55
.035	0.88		.200	5.08
.050	1.27		.770	19.55

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Relay widths are given in table I.
4. A position identifier (shape optional) shall be placed on the side of each relay to designate pin 1.
5. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

© FIGURE 1. Outline drawing - Continued.

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TABLE I. Dash numbers and applicable characteristics.

Dash number			Circuit diagram (see fig 1.)	DC coil data at 25°C												Relay width (maximum) inch
				Resistance (minimum) ohms			Pickup voltage (maximum)			Dropout voltage (minimum)			Power dissipation (nominal) mW			
Coil voltage				Coil voltage			Coil voltage			Coil voltage						
5	12	24		5	12	24	5	12	24	5	12	24	5	12	24	
001	002	003	A	180	450	1,835	3.7	8.0	18.0	0.5	1.0	2.0	125	288	270	.290
004	005	006	B	"	"	"	"	"	"	"	"	"	"	"	"	.290
007	008	009	C	"	"	"	"	"	"	"	"	"	"	"	"	.350
010	011	012	D	"	"	"	"	"	"	"	"	"	"	"	"	.350

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REQUIREMENTS:

DESIGN AND CONSTRUCTION:

Dimensions and configuration: See figure 1.

Diode characteristics (when applicable): Diodes shall be JANTX as a minimum. 1/

Peak inverse voltage: 75.0 V dc.

CONTACT DATA:

Arrangements: See figure 1.

Load ratings:

Contacts	V dc (max)	mA (max)	VA (max)
Form A	100	500	10
Form B	100	500	10

Contact resistance:

Before life: 0.130 ohm maximum.

During and after life: 1.0 ohm maximum.

Group A inspection: 0.200 ohm maximum.

Contact noise: Applicable.

Contact stability: Applicable.

COIL DATA: See table I.

Duty rating: Continuous.

Coil power dissipation: See table I.

Operate time (including bounce): See table II.

Release time (including bounce): See table II.

TABLE II. Operate and release times (including bounce).

Contact arrangement	Operate time (nominal)		Release time (nominal)	
	Without diode	With diode	Without diode	With diode
	ms	ms	ms	ms
Form A	1	1	.5	1
Form B	.5	.5	.5	.5

1/ WARNING: Reverse polarity on coil terminals will destroy diode.

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ELECTRICAL DATA:

Dielectric withstanding voltage: MIL-STD-202, method 301. 2/

Test points:

Between all mutually insulated components: 500 V dc.

Across contacts: 250 V dc.

Insulation resistance: 10,000 MΩ *minimum.*

Test potential: 100 V dc ±10%.

Test points: Between contacts.

Coil transient suppression (relays with internal diodes): 1.0 V maximum negative transient.

Capacitance, pF maximum:

Across open contacts:	1.25
Coil to contacts:	3.0

Test points: Coil to contacts.

ENVIRONMENTAL DATA:

Thermal EMF: Not applicable.

Thermal shock: Applicable.

Shock (specified pulse): Applicable.

Vibration, high frequency: Applicable.

Salt spray (corrosion): Applicable.

Moisture resistance: Applicable.

Magnetic interference: Single in-line packages can be mounted side-by-side in the same socket. This mounting will cause their magnetic field to interact with each other, affecting pickup and dropout voltages by as much as 15 percent of original parameters. Relays are also subject to interaction with stray magnetic fields. To keep within operating parameters, relays should be mounted no closer than .500 inch (12.7 mm) of each other.

Resistance to soldering heat: Applicable.

PHYSICAL:

Enclosure: Molded plastic.

Operating temperature range: -40°C to +105°C.

Lead integrity: MIL-STD-883, method 2004, test condition B2; the force to be applied shall be 8.0 ±0.5 ounces.

Solderability: Applicable.

2/ For relays supplied with an internal diode, the coil leads shall be connected together to avoid damage to the diode.

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Life: Applicable.

Weight: 3.3 grams maximum.

Part or Identifying Number: M83516/3-(dash number from table I).

QUALIFICATION:

Qualification inspection and sample size: See table III.

TABLE III. Qualification inspection and sample size.

M83516/3-	Number of samples
005 011	13 units each part number. Qualification inspection as applicable.
004 and 006 010 and 012	2 units each part number. Qualification inspection group I and life.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F715-01)

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